

MRUNAL PRADEEP PATIL

AI-Focused Software Engineer | 3+ Years Experience | Software Engineering Graduate

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AI-Focused Software Engineer and Computer Science Graduate combining 3+ years Full-Stack Software development expertise with cutting-edge LLM implementations. Specialized in Java, Microservices, Spring Boot, AWS, Database Management, DevOps, and React, while mastering AI-powered automation using LLaMA 3, LangChain, and NLP for secure on-premise data processing. Demonstrated track record of designing and implementing solutions that seamlessly integrate modern software engineering practices with cutting-edge technologies.

EDUCATION & CERTIFICATION

Master of Science – Software Engineering

George Mason University, Fairfax, Virginia

Jan 2024 - Dec 2025

GPA: 3.80/4.00

Bachelor of Technology - Computer Science and Engineering

SRM Institute of Science and Technology, Chennai, India

Aug 2017 - May 2021

GPA: 3.74/4.00

Relevant Coursework: Algorithm Design and Analysis, Advanced Java Programming, Advanced Data Structures, Database Systems, Cloud Computing, Object Oriented Software Specifications, Artificial Intelligence, Machine Learning & Deep Learning.

Certification : [AWS Certified Developer Associate \(DVA-C02\)](#)

PROFESSIONAL EXPERIENCE

Graduate Research Assistant | Virtual Library of Virginia, George Mason University, Fairfax, Virginia

Jan 2025 – Present

- Designed a secure, on-premise AI-powered data standardization system, **reducing manual work time by 95%** and ensuring 100% compliance with strict **data privacy** policies.
- Implemented transformation pipelines using **local LLaMA 3 models (via Ollama)** to interpret PDF/XLSX rule files, with intelligent **fallback** to deterministic mappings to maintain accuracy even when AI is uncertain.
- Build an end-to-end workflow integrating React, Spring Boot, and FastAPI, enabling simple file uploads, automated AI-based transformations and instant downloadable reports.
- Build an AI-driven contract analysis tool using **spaCy and Hugging Face Transformers** to extract relevant licensing terms from legal agreements, **cutting review time by 70%** and improving compliance consistency.
- Developed an AI-assisted metadata extraction workflow for Hyku repository, using PyMuPDF/pdfplumber and LLaMA 3 to summarize presentations, **reducing manual entry time by 80%** and increasing metadata completeness.

Software Engineer | Cognizant Technology Services, Bangalore, India

Aug 2021 – Sep 2023

- Contributed to the development of new user interface features for Credit Card's fraud reporting system using **React**, improving user engagement by **25%** through better **performance** and **usability**.
- Worked on migration of legacy UI applications to React, improving **application performance by over 35%**, increasing **scalability by 30%**, and significantly enhancing code **maintainability** and **developer efficiency**.
- Managed and maintained backend systems using **Java and Spring Boot**, achieving a **99.9% uptime** and ensuring seamless integration with frontend applications improving system reliability.
- Designed and implemented new **RESTful API** endpoints, expanding system capabilities, and supporting evolving business requirements, which contributed to increase in operational efficiency.
- Significantly enhanced the performance of existing products by incorporating caching mechanisms and **optimizing SQL queries** and code logic, resulting in a remarkable **reduction of operation time by over 40%**, significantly improving user experience.
- Followed Test Driven Development by writing Unit tests and Integration tests with **test coverage above 95%**.

Software Engineer Intern | Cognizant Technology Services, Bangalore, India

Jan 2021 – Jun 2021

- Worked on transition of an internal **monolithic billing application to a microservices architecture** using Java, Spring Boot, enhancing modularity, scalability and maintainability.

TECHNICAL SKILLS

Languages & Databases: C, Java, JavaScript, TypeScript, Python, MySQL, MongoDB, DynamoDB

AI & Machine Learning: PyTorch, scikit-learn, Hugging Face, LangChain, Large Language Models, LLaMA 3, spaCy, Prompt Engineering

AI Development Tools: Cursor AI, GitHub CoPilot, Tabnine

Backend & Data Library: FastAPI, Spring MVC, Spring Boot, JPA, Hibernate, Pandas, PyMuPDF, pdfplumber

Cloud: AWS - S3, EC2, Lambda, SNS, SQS, RDS, Step Functions, API Gateway

Testing Frameworks: JUnit, Mockito, Playwright, Jest

DevOps-CI/CD & Monitoring: Docker, Jenkins, Kubernetes, Kibana, CloudWatch

ACADEMIC PROJECTS

AI-Powered Healthcare Appointment (React, Python, Flask, scikit-learn, ML, Docker, AWS, Kubernetes)

Spring 2025

- Developed and trained a custom ML model (Naive Bayes with scikit-learn) and deployed it via a Flask REST API to predict doctor specialties based on user symptoms. Deployed the full-stack app using Docker and Kubernetes on AWS for scalability.

SmartShop-AI-Powered Web App (React, Java, Spring Boot, PostgreSQL, JWT, OpenAI API)

Winter 2024

- Build a secure app with role-based access using JWT and integrated OpenAI API to auto generate product descriptions and enhance search intelligence.